

5×5 grid: HILP versus prun-HILP (E+F)

$\alpha = 1$; λ is the retained fraction. Fixed duration = 1; Manhattan heuristic. 3 search trials per complete configuration; median wall time includes filtering, model construction and optimization.

Underlined time: lower than the recorded HILP baseline. **Bold Obj.**: differs from HILP; lower cost is better. RAO* uses historical measurements and a different native objective boundary.

Table 1: Simulation results with heuristics: prun-HILP (E+F).

h	Δ	HILP				RAO* (historical)				prun-HILP (E+F) ($\lambda = 0.3$)				prun-HILP (E+F) ($\lambda = 0.5$)				prun-HILP (E+F) ($\lambda = 0.7$)				prun-HILP (E+F) ($\lambda = 0.9$)			
		Obj.	Time (s)	n	Iter.	Obj.	Time (s)	n	Iter.	Obj.	Time (s)	n	Iter.	Obj.	Time (s)	n	Iter.	Obj.	Time (s)	n	Iter.	Obj.	Time (s)	n	Iter.
3	0.1	8.9323	0.049	268	12	9.7497	0.080	337	40	9.8556	<u>0.022</u>	52	4	9.7377	<u>0.031</u>	64	6	9.0052	0.051	196	10	9.0000	0.064	268	12
	0.2	8.1556	0.040	160	9	8.2410	0.028	142	13	9.8556	<u>0.021</u>	52	4	9.9050	<u>0.029</u>	64	6	9.6735	0.043	88	9	8.1556	0.050	160	9
	0.3	8.1556	0.041	172	9	8.2410	0.028	175	15	9.8556	<u>0.021</u>	52	4	9.8556	<u>0.021</u>	52	4	9.6787	<u>0.030</u>	76	6	8.1556	0.048	172	9
4	0.1	9.6798	0.194	1084	22	10.7752	0.530	1309	135	10.7910	<u>0.039</u>	160	5	10.6811	<u>0.063</u>	196	8	9.9591	0.216	844	20	9.8562	0.354	1180	25
	0.2	8.2410	0.111	592	13	8.3305	0.193	550	48	10.7910	<u>0.042</u>	160	5	10.8400	<u>0.062</u>	196	8	10.5874	<u>0.100</u>	328	10	8.7916	0.148	568	13
	0.3	8.2410	0.150	892	20	8.3305	0.209	712	60	10.7910	<u>0.038</u>	160	5	10.7862	<u>0.059</u>	208	8	10.7117	<u>0.090</u>	328	10	8.2410	0.278	868	21
5	0.1	10.3580	1.823	4504	45	11.1940	3.276	5170	470	11.7295	<u>0.130</u>	556	9	11.6254	<u>0.184</u>	724	11	11.1642	0.622	2236	25	10.4272	<u>0.713</u>	2524	26
	0.2	8.3305	0.499	3004	26	8.5173	1.998	3073	259	11.7295	<u>0.139</u>	556	9	11.7777	<u>0.188</u>	724	12	11.4595	<u>0.380</u>	1384	17	9.3362	1.208	3508	32
	0.3	8.3305	0.633	3880	29	8.5173	2.701	4228	353	11.7295	<u>0.160</u>	556	9	11.7285	<u>0.203</u>	772	12	11.6102	<u>0.328</u>	1168	15	8.7677	0.947	2860	25
6	0.1	10.8564	7.654	13048	63	11.6643	33.628	25393	2196	12.6733	<u>0.416</u>	1864	15	12.5734	<u>0.760</u>	2548	18	12.6557	<u>2.448</u>	7732	33	11.1629	<u>5.888</u>	10624	46
	0.2	8.5173	5.109	17608	49	9.4548	54.247	48844	4100	12.6733	<u>0.419</u>	1864	15	12.7170	<u>0.728</u>	2512	18	12.3763	<u>1.660</u>	5044	25	9.4554	<u>7.083</u>	15088	48
	0.3	8.5173	9.442	31048	76	9.4548	58.761	50839	4256	12.6733	<u>0.426</u>	1864	15	12.6614	<u>0.653</u>	2560	17	12.4962	<u>1.644</u>	5152	24	8.9082	<u>6.293</u>	13528	47

DARP $n = |E \cup F|$ counts generated action histories; RAO* n counts belief graph nodes. Iter. counts p-ILP calls for DARP and expansions for RAO*. An executable pruned policy does not certify global optimality; failed groups are –.

Pruning diagnostics.

λ	Success/runs	Speedup	Max. cost increase (%)	E sent (%)	F sent (%)	Filter (s)	Fallbacks
0.3	36/36	<u>3.97×</u>	+48.80	99.4	32.5	0.007	0
0.5	36/36	<u>2.86×</u>	+49.31	94.9	51.9	0.012	0
0.7	36/36	<u>1.51×</u>	+46.72	83.9	72.7	0.041	0
0.9	36/36	<u>0.75×</u>	+12.07	93.3	91.2	0.088	0

Cost increase is $100(C_{\text{pruned}} - C_{\text{HILP}})/|C_{\text{HILP}}|$; the diagnostic column reports the largest configuration-level increase, using median costs. Speedup is the median configuration-level time ratio. E/F sent ratios include fallback submissions.

F-only skips E-subtree pruning; necessary dead-end, prefix-risk and unreachable-history filtering still applies. Submitted coefficients and global risk constraints are preserved.

HILP and E+F reuse the completed initial batch; F-only was added in a supplementary serial batch. Underlining shows observed differences, not statistical significance. MIPGap = 10^{-6} ; search limit = 120 s. Replay (1000 episodes for trial 1) is outside search timing.

5 × 5 grid: HILP versus prun-HILP (F-only)

$\alpha = 1$; λ is the retained fraction. Fixed duration = 1; Manhattan heuristic. 3 search trials per complete configuration; median wall time includes filtering, model construction and optimization. Underlined time: lower than the recorded HILP baseline. **Bold Obj.**: differs from HILP; lower cost is better. RAO* uses historical measurements and a different native objective boundary.

Table 2: Simulation results with heuristics: prun-HILP (F-only).

h	Δ	HILP				RAO* (historical)				prun-HILP (F-only) ($\lambda = 0.3$)				prun-HILP (F-only) ($\lambda = 0.5$)				prun-HILP (F-only) ($\lambda = 0.7$)				prun-HILP (F-only) ($\lambda = 0.9$)			
		Obj.	Time (s)	n	Iter.	Obj.	Time (s)	n	Iter.	Obj.	Time (s)	n	Iter.	Obj.	Time (s)	n	Iter.	Obj.	Time (s)	n	Iter.	Obj.	Time (s)	n	Iter.
3	0.1	8.9323	0.049	268	12	9.7497	0.080	337	40	9.8556	<u>0.022</u>	52	4	9.7377	<u>0.030</u>	64	5	8.9323	<u>0.049</u>	196	10	8.9323	0.059	268	12
	0.2	8.1556	0.040	160	9	8.2410	0.028	142	13	9.8556	<u>0.021</u>	52	4	9.9050	<u>0.027</u>	64	5	9.6735	<u>0.032</u>	88	7	8.1556	0.047	160	9
	0.3	8.1556	0.041	172	9	8.2410	0.028	175	15	9.8556	<u>0.022</u>	52	4	9.8556	<u>0.022</u>	52	4	9.6787	<u>0.029</u>	76	6	8.1556	0.048	172	9
4	0.1	9.6798	0.194	1084	22	10.7752	0.530	1309	135	10.7910	<u>0.038</u>	160	5	10.6811	<u>0.064</u>	196	8	9.7775	<u>0.126</u>	472	14	9.6798	0.307	1084	22
	0.2	8.2410	0.111	592	13	8.3305	0.193	550	48	10.7910	<u>0.037</u>	160	5	10.8400	<u>0.062</u>	196	8	10.5874	<u>0.090</u>	328	10	8.7916	0.147	568	13
	0.3	8.2410	0.150	892	20	8.3305	0.209	712	60	10.7910	<u>0.038</u>	160	5	10.7862	<u>0.059</u>	208	8	10.7117	<u>0.098</u>	328	10	8.2410	0.235	868	20
5	0.1	10.3580	1.823	4504	45	11.1940	3.276	5170	470	11.7295	<u>0.138</u>	556	9	11.6254	<u>0.177</u>	724	11	10.6402	<u>0.492</u>	1492	23	10.4252	<u>0.834</u>	2692	28
	0.2	8.3305	0.499	3004	26	8.5173	1.998	3073	259	11.7295	<u>0.171</u>	556	9	11.7777	<u>0.215</u>	724	12	11.4735	<u>0.328</u>	1132	13	9.3362	0.772	2824	24
	0.3	8.3305	0.633	3880	29	8.5173	2.701	4228	353	11.7295	<u>0.137</u>	556	9	11.7285	<u>0.192</u>	772	12	11.6416	<u>0.307</u>	1156	14	8.7677	0.841	2860	25
6	0.1	10.8564	7.654	13048	63	11.6643	33.628	25393	2196	12.6733	<u>0.453</u>	1864	15	12.5757	<u>0.576</u>	2416	15	11.6273	2.243	5284	33	11.1126	<u>4.697</u>	9124	43
	0.2	8.5173	5.109	17608	49	9.4548	54.247	48844	4100	12.6733	<u>0.461</u>	1864	15	12.7193	<u>0.604</u>	2380	16	12.4123	<u>1.423</u>	4876	23	9.8348	<u>5.087</u>	12712	40
	0.3	8.5173	9.442	31048	76	9.4548	58.761	50839	4256	12.6733	<u>0.397</u>	1864	15	12.6688	<u>0.604</u>	2476	15	12.5767	<u>1.173</u>	4348	21	9.2977	<u>6.251</u>	13408	48

DARP $n = |E \cup F|$ counts generated action histories; RAO* n counts belief graph nodes. Iter. counts p-ILP calls for DARP and expansions for RAO*. An executable pruned policy does not certify global optimality; failed groups are –.

Pruning diagnostics.

λ	Success/runs	Speedup	Max. cost increase (%)	E sent (%)	F sent (%)	Filter (s)	Fallbacks
0.3	36/36	<u>4.29×</u>	+48.80	100.0	32.4	0.005	0
0.5	36/36	<u>2.79×</u>	+49.33	100.0	52.0	0.010	0
0.7	36/36	<u>1.54×</u>	+47.66	100.0	73.3	0.021	0
0.9	36/36	0.84×	+15.47	100.0	91.6	0.071	0

Cost increase is $100(C_{\text{pruned}} - C_{\text{HILP}})/|C_{\text{HILP}}|$; the diagnostic column reports the largest configuration-level increase, using median costs. Speedup is the median configuration-level time ratio. E/F sent ratios include fallback submissions.

F-only skips E-subtree pruning; necessary dead-end, prefix-risk and unreachable-history filtering still applies. Submitted coefficients and global risk constraints are preserved.

HILP and E+F reuse the completed initial batch; F-only was added in a supplementary serial batch. Underlining shows observed differences, not statistical significance. MIPGap = 10^{-6} ; search limit = 120 s. Replay (1000 episodes for trial 1) is outside search timing.

100×100 grid: HILP versus prun-HILP (E+F)

$\alpha = 1$; λ is the retained fraction. Fixed duration = 1; Manhattan heuristic. 3 search trials per complete configuration; median wall time includes filtering, model construction and optimization. Underlined time: lower than the recorded HILP baseline. **Bold Obj.**: differs from HILP; lower cost is better. RAO* uses historical measurements and a different native objective boundary.

Table 3: Simulation results with heuristics: prun-HILP (E+F).

h	Δ	HILP				RAO* (historical)				prun-HILP (E+F) ($\lambda = 0.3$)				prun-HILP (E+F) ($\lambda = 0.5$)				prun-HILP (E+F) ($\lambda = 0.7$)				prun-HILP (E+F) ($\lambda = 0.9$)			
		Obj.	Time (s)	n	Iter.	Obj.	Time (s)	n	Iter.	Obj.	Time (s)	n	Iter.	Obj.	Time (s)	n	Iter.	Obj.	Time (s)	n	Iter.	Obj.	Time (s)	n	Iter.
3	0.1	198.9323	0.047	268	12	199.7497	0.082	337	40	199.8556	<u>0.022</u>	52	4	199.7377	<u>0.030</u>	64	6	199.0052	0.052	196	10	199.0000	0.064	268	12
	0.2	198.1556	0.044	160	9	198.2410	0.029	142	13	199.8556	<u>0.023</u>	52	4	199.9050	<u>0.037</u>	64	6	199.6735	<u>0.043</u>	88	8	198.1556	0.052	160	9
	0.3	198.1556	0.042	172	9	198.2410	0.031	175	15	199.8556	<u>0.022</u>	52	4	199.8556	<u>0.022</u>	52	4	199.6787	<u>0.031</u>	76	6	198.1556	0.051	172	9
4	0.1	199.6798	0.226	1084	22	200.7752	0.512	1309	135	200.7910	<u>0.039</u>	160	5	200.6811	<u>0.066</u>	196	8	199.9591	<u>0.225</u>	844	20	199.8512	0.371	1168	24
	0.2	198.2410	0.115	592	13	198.3305	0.203	550	48	200.7910	<u>0.039</u>	160	5	200.8400	<u>0.065</u>	196	8	200.5871	<u>0.088</u>	304	9	198.7916	0.140	544	13
	0.3	198.2410	0.154	892	20	198.3305	0.216	712	60	200.7910	<u>0.038</u>	160	5	200.7862	<u>0.060</u>	208	8	200.7117	<u>0.097</u>	328	10	198.7916	0.216	844	18
5	0.1	200.3580	1.729	4504	45	201.1839	3.168	4978	455	201.7295	<u>0.126</u>	544	9	201.6254	<u>0.185</u>	712	11	201.1744	<u>0.570</u>	2056	25	200.4259	<u>0.707</u>	2512	26
	0.2	198.3305	0.506	2980	26	198.4236	1.441	2104	178	201.7295	<u>0.119</u>	544	9	201.7777	<u>0.179</u>	724	12	201.4727	<u>0.415</u>	1348	19	199.7273	1.588	3940	37
	0.3	198.3305	0.577	3880	29	198.4236	1.745	2704	226	201.7295	<u>0.119</u>	544	9	201.7285	<u>0.176</u>	772	12	201.6054	<u>0.328</u>	1228	16	199.6580	1.890	4732	41
6	0.1	200.8523	5.018	12172	54	201.6259	22.403	15550	1334	202.6732	<u>0.364</u>	1732	12	202.5734	<u>0.651</u>	2488	16	202.5570	<u>2.365</u>	7144	34	201.1683	12.270	12208	75
	0.2	198.4236	2.841	13864	39	199.3486	36.773	28735	2419	202.6732	<u>0.363</u>	1732	12	202.7176	<u>0.666</u>	2464	17	202.3829	<u>1.647</u>	4708	25	199.9435	4.474	12016	41
	0.3	198.4236	3.748	17260	53	198.8046	12.678	7477	638	202.6732	<u>0.357</u>	1732	12	202.6686	<u>0.558</u>	2536	14	202.4762	<u>1.794</u>	4984	25	199.7558	5.993	12856	48

DARP $n = |E \cup F|$ counts generated action histories; RAO* n counts belief graph nodes. Iter. counts p-ILP calls for DARP and expansions for RAO*. An executable pruned policy does not certify global optimality; failed groups are –.

Pruning diagnostics.

λ	Success/runs	Speedup	Max. cost increase (%)	E sent (%)	F sent (%)	Filter (s)	Fallbacks
0.3	36/36	<u>4.55×</u>	+2.14	100.0	32.8	0.007	0
0.5	36/36	<u>3.05×</u>	+2.16	97.1	51.9	0.012	0
0.7	36/36	<u>1.46×</u>	+2.04	84.0	72.7	0.050	0
0.9	36/36	0.67×	+0.77	91.3	91.2	0.097	0

Cost increase is $100(C_{\text{pruned}} - C_{\text{HILP}})/|C_{\text{HILP}}|$; the diagnostic column reports the largest configuration-level increase, using median costs. Speedup is the median configuration-level time ratio. E/F sent ratios include fallback submissions.

F-only skips E-subtree pruning; necessary dead-end, prefix-risk and unreachable-history filtering still applies. Submitted coefficients and global risk constraints are preserved.

HILP and E+F reuse the completed initial batch; F-only was added in a supplementary serial batch. Underlining shows observed differences, not statistical significance. MIPGap = 10^{-6} ; search limit = 120 s. Replay (1000 episodes for trial 1) is outside search timing.

100 × 100 grid: HILP versus prun-HILP (F-only)

$\alpha = 1$; λ is the retained fraction. Fixed duration = 1; Manhattan heuristic. 3 search trials per complete configuration; median wall time includes filtering, model construction and optimization. Underlined time: lower than the recorded HILP baseline. **Bold Obj.**: differs from HILP; lower cost is better. RAO* uses historical measurements and a different native objective boundary.

Table 4: Simulation results with heuristics: prun-HILP (F-only).

h	Δ	HILP				RAO* (historical)				prun-HILP (F-only) ($\lambda = 0.3$)				prun-HILP (F-only) ($\lambda = 0.5$)				prun-HILP (F-only) ($\lambda = 0.7$)				prun-HILP (F-only) ($\lambda = 0.9$)			
		Obj.	Time (s)	n	Iter.	Obj.	Time (s)	n	Iter.	Obj.	Time (s)	n	Iter.	Obj.	Time (s)	n	Iter.	Obj.	Time (s)	n	Iter.	Obj.	Time (s)	n	Iter.
3	0.1	198.9323	0.047	268	12	199.7497	0.082	337	40	199.8556	<u>0.022</u>	52	4	199.7377	<u>0.029</u>	64	5	198.9323	0.053	196	10	198.9323	0.063	268	12
	0.2	198.1556	0.044	160	9	198.2410	0.029	142	13	199.8556	<u>0.021</u>	52	4	199.9050	<u>0.027</u>	64	5	199.6735	<u>0.033</u>	88	7	198.1556	0.047	160	9
	0.3	198.1556	0.042	172	9	198.2410	0.031	175	15	199.8556	<u>0.022</u>	52	4	199.8556	<u>0.022</u>	52	4	199.6787	<u>0.031</u>	76	6	198.1556	0.052	172	9
4	0.1	199.6798	0.226	1084	22	200.7752	0.512	1309	135	200.7910	<u>0.042</u>	160	5	200.6811	<u>0.076</u>	196	8	199.7775	<u>0.165</u>	472	14	199.6798	0.335	1084	22
	0.2	198.2410	0.115	592	13	198.3305	0.203	550	48	200.7910	<u>0.040</u>	160	5	200.8400	<u>0.066</u>	196	8	200.5871	<u>0.090</u>	304	9	198.7916	0.141	544	13
	0.3	198.2410	0.154	892	20	198.3305	0.216	712	60	200.7910	<u>0.041</u>	160	5	200.7862	<u>0.063</u>	208	8	200.7117	<u>0.094</u>	328	10	198.7916	0.228	844	17
5	0.1	200.3580	1.729	4504	45	201.1839	3.168	4978	455	201.7295	<u>0.127</u>	544	9	201.6254	<u>0.181</u>	712	11	200.6833	<u>0.490</u>	1624	22	200.4252	<u>0.949</u>	2692	28
	0.2	198.3305	0.506	2980	26	198.4236	1.441	2104	178	201.7295	<u>0.122</u>	544	9	201.7777	<u>0.185</u>	724	12	201.4886	<u>0.418</u>	1312	20	199.6468	0.681	2752	23
	0.3	198.3305	0.577	3880	29	198.4236	1.745	2704	226	201.7295	<u>0.122</u>	544	9	201.7285	<u>0.188</u>	772	12	201.6505	<u>0.349</u>	1204	16	199.6593	1.266	3580	32
6	0.1	200.8523	5.018	12172	54	201.6259	22.403	15550	1334	202.6732	<u>0.389</u>	1732	12	202.5756	<u>0.664</u>	2404	15	201.6617	<u>1.756</u>	5056	29	201.1355	9.213	9784	59
	0.2	198.4236	2.841	13864	39	199.3486	36.773	28735	2419	202.6732	<u>0.408</u>	1732	12	202.7198	<u>0.657</u>	2356	16	202.4242	<u>1.428</u>	4540	24	200.3049	4.431	10108	44
	0.3	198.4236	3.748	17260	53	198.8046	12.678	7477	638	202.6732	<u>0.337</u>	1732	12	202.6699	<u>0.534</u>	2488	14	202.5751	<u>1.235</u>	4240	21	200.2691	6.045	12364	50

DARP $n = |E \cup F|$ counts generated action histories; RAO* n counts belief graph nodes. Iter. counts p-ILP calls for DARP and expansions for RAO*. An executable pruned policy does not certify global optimality; failed groups are –.

Pruning diagnostics.

λ	Success/runs	Speedup	Max. cost increase (%)	E sent (%)	F sent (%)	Filter (s)	Fallbacks
0.3	36/36	<u>4.43×</u>	+2.14	100.0	32.8	0.006	0
0.5	36/36	<u>2.85×</u>	+2.17	100.0	52.0	0.010	0
0.7	36/36	<u>1.50×</u>	+2.09	100.0	73.0	0.041	0
0.9	36/36	<u>0.71×</u>	+0.95	100.0	91.4	0.059	0

Cost increase is $100(C_{\text{pruned}} - C_{\text{HILP}})/|C_{\text{HILP}}|$; the diagnostic column reports the largest configuration-level increase, using median costs. Speedup is the median configuration-level time ratio. E/F sent ratios include fallback submissions.

F-only skips E-subtree pruning; necessary dead-end, prefix-risk and unreachable-history filtering still applies. Submitted coefficients and global risk constraints are preserved.

HILP and E+F reuse the completed initial batch; F-only was added in a supplementary serial batch. Underlining shows observed differences, not statistical significance. MIPGap = 10^{-6} ; search limit = 120 s. Replay (1000 episodes for trial 1) is outside search timing.